

CE310 — Engineering Systems Analysis

Pre-Lab Quiz: Week 1 — The Flaw of Averages

Due: Before Week 1 Lab (Class 3) — submit via D2L Total: 10 points (1 point each)
Estimated time: ~10 minutes

Instructions: Select the single best answer for each question. This quiz is auto-graded and must be completed before attending the Week 1 lab session. You may use your notes and readings, but work independently.

Questions

Q1. When the U.S. Air Force measured 10 physical dimensions of thousands of pilots to design a cockpit for the “average” pilot, how many pilots were average across all 10 dimensions simultaneously?

- A) Fewer than 100
 - B) About 5% of all pilots
 - C) About 1% of all pilots
 - D) Zero
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Q2. A weather station records one temperature measurement per hour for a full calendar year (non-leap year). How many data rows does the dataset contain?

- A) 365
 - B) 4,380
 - C) 8,760
 - D) 52,560
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Q3. An HVAC engineer sizes a cooling system based solely on the annual average outdoor temperature. On the hottest day of the year, what is the most likely outcome?

- A) The system will be oversized and waste energy
 - B) The system will perform exactly as designed
 - C) The system will be undersized and fail to maintain comfortable indoor temperatures
 - D) The building will automatically switch to a backup system
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Q4. ASHRAE Standard 169 defines the cooling design dry-bulb temperature as the value corresponding to the hottest 1% of annual hours. For an 8,760-hour year, this is calculated as the:

- A) Single highest hourly temperature recorded
 - B) Average of the top 100 hourly temperatures
 - C) 876th-highest hourly temperature
 - D) 88th-highest hourly temperature
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Q5. Which of the following best describes what standard deviation measures?

- A) The difference between the maximum and minimum values
 - B) The middle value when data is sorted from smallest to largest
 - C) How much individual values typically deviate from the mean
 - D) The most frequently occurring value in a dataset
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Q6. A weather station records a maximum annual temperature of 112 °F and a minimum of 28 °F. What is the annual temperature range?

- A) 56 °F
 - B) 70 °F
 - C) 84 °F
 - D) 140 °F
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Q7. A structural engineer designs a roadside billboard using the average annual wind speed to calculate design wind pressure. Compared to the actual peak wind speed, this approach is likely to:

- A) Overestimate wind pressure, making the design unnecessarily expensive
 - B) Accurately represent the worst-case wind load the billboard will experience
 - C) Have no effect, since wind pressure depends on factors other than wind speed
 - D) Underestimate wind pressure, risking structural failure during a storm
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Q8. When calculating the standard deviation of a sample (rather than an entire population), why do you divide by $(n - 1)$ instead of n ?

- A) It only matters for very large datasets
 - B) It corrects for a bias that causes sample-based estimates to underestimate the true population variability
 - C) It converts standard deviation into variance
 - D) It is a rounding convention with no mathematical basis
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Q9. Which Excel formula returns the 88th-largest value in the range B2:B8761?

- A) =MAX(B2:B8761, 88)
- B) =RANK(B2:B8761, 88)

- C) =PERCENTILE(B2:B8761, 88)
 - D) =LARGE(B2:B8761, 88)
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Q10. A student wants to count how many hours in the dataset had a temperature greater than 100 °F. Which Excel formula is correct?

- A) =COUNT(B2:B8761, ">100")
- B) =SUMIF(B2:B8761, ">100")
- C) =COUNTIF(B2:B8761, ">100")
- D) =FILTER(B2:B8761, ">100")